ISLAMIC UNIVERSITY OF GAZA

FACULTY OF ENGINEERING

ARTIFICIAL INTELLIGENCE ENGINEERING

TECHNICAL REPORT: SMART INFASTRUCTURE TRAFFIC CONTROL SYSTEM

Design & Implementation of a 16-State Finite State Machine with Multimodal Isolated Logic

MODULO-16 SYNCHRONOUS FSM LOOP

STATES S0–S15

ABSOLUTE PEDESTRIAN ISOLATION ENFORCED

Final Project

- FINITE STATE MACHINE CONTROLLER
- OPTIMIZED BOOLEAN EQUATIONS (SOP)
- 75% GATE REDUNDANCY REDUCTION (RIGHT TURNS)



Course Name & code: Digital Logic Design (ECOM1101)
Title: Logic Gates
Submitted to: Eng Hashem
Date of Submission: 12/June/2026

Prepared By: Eng Hamza Shaban
Student ID: 120257122

Objectives:

Design and Implementation of a 16-State Finite State Machine Traffic Controller with Advanced Pedestrian and Left-Turn Isolation Logic.

EXECUTIVE SUMMARY:

This technical report documents the complete design, mathematical optimization, and schematic verification of an advanced 4-way traffic intersection controller. Operating as a synchronous Modulo-16 Finite State Machine (FSM), the system controls sixteen distinct outputs managing forward traffic, dedicated left-turn corridors, and pedestrian crossings across North, South, East, and West boundaries. By translating empirical traffic flow requirements into rigorous Boolean algebraic models, the final circuit schema achieves a 75% reduction in gate redundancy for directional turn-subsystems through synchronized term-sharing. The entire system has been structurally simulated and verified within Logisim environments to serve as a high-end engineering portfolio asset.

SYSTEM SPECIFICATION & FSM STATE ARCHITECTURE:

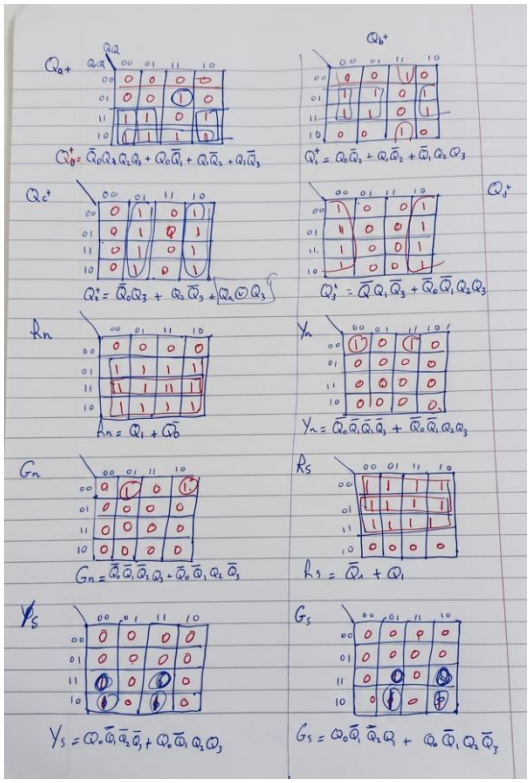
The intersection is modeled as a synchronous state-dependent system divided into four macro-phases (North, East, South, West). Each macro-phase possesses a 4-state cycle: Forward Green, Forward Yellow, Left-Turn Green, and Left-Turn Yellow. Pedestrian crossing segments are structurally interleaved into the state mapping to enforce absolute physical isolation from conflicting vehicular vectors.

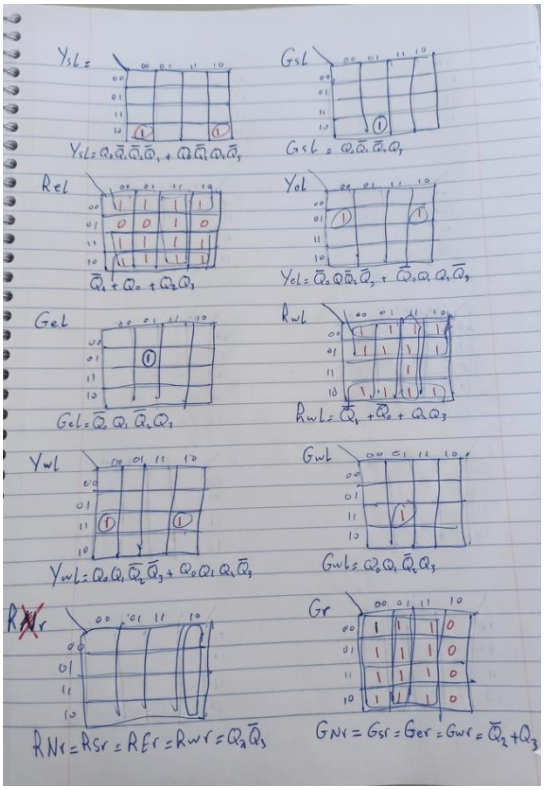
The system relies on four state variables (Q3, Q2, Q1, Q0) yielding exactly sixteen discrete binary combinations (S0 through S15). The sequence progresses deterministically on each positive edge clock pulse according to the state transition matrix:

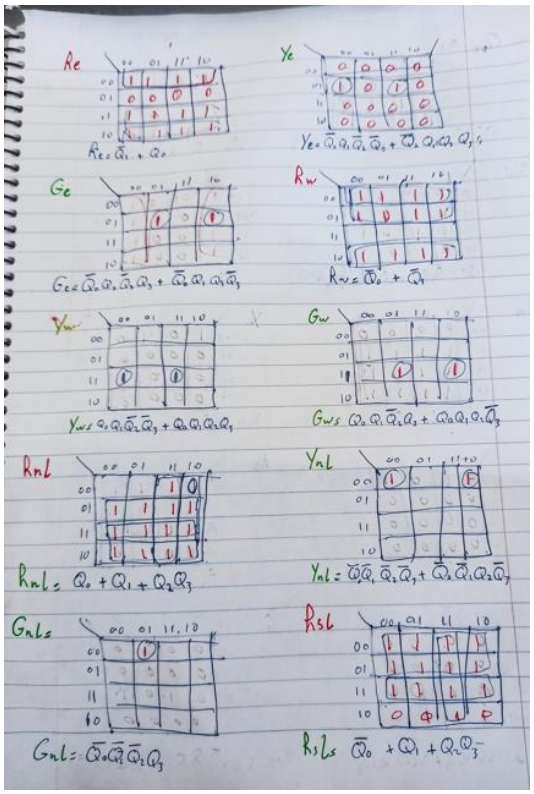
Macro-Axis	State Label	Binary State ($Q_3Q_2Q_1Q_0$)	Active Vehicular Corridor
North Bound	S0	0000	North (Yellow) & North left (Yellow) & All Pedestrian Isolation Phase & All Rigths open
	S1	0001	North(Green) & Norht left (Green) & All Pedestrian Isolation Phase & All Rigths open
	S2	0010	North (Green) & North Left (Yellow) & PedestrianEast&West (opened) & All Right closed
	S3	0011	North(Yellow) & North Left (Red) & All Pedestrian Isolation Phase & All Rigths open
East Bound	S4	0100	East (Yellow) & East Left (Yellow) & All Pedestrian Isolation Phase & All Rigths open
	S5	0101	East (Green) & East Left(Green) & All Pedestrian Isolation Phase & All Rigths open
	S6	0110	East (Green) & East Left (Yellow) & PedestrianNorth&South(opened) & All Right closed
	S7	0111	East(Yellow) & East Left(Red) & All Pedestrian Isolation Phase & All Rigths open
South Bound	S8	1000	South (Yellow) & South Left (Yellow) & All Pedestrian Isolation Phase & All Rigths open
	S9	1001	South (Green) & South Left (Green) & All Pedestrian Isolation Phase & All Rigths open
	S10	1010	South(Green) & South Left (Yellow) & PedestrianEast & West (opened) & All Right closed
	S11	1011	South (Yellow) & South Left(Red) & All Pedestrian Isolation Phase & All Rigths open
West Bound	S12	1100	West (Yellow) & West Left (Yellow) & All Pedestrian Isolation Phase & All Rigths open
	S13	1101	West(Green) & West Left (Green) & All Pedestrian Isolation Phase & All Rigths open
	S14	1110	West(Green) & West Left(Yellow) & PedestrianNorth&South (opened) & All Right closed
	S15	1111	West(Yellow) & West Left(Red) & All Pedestrian Isolation Phase & All Rigths open

COMBINATIONAL LOGIC OPTIMIZATION & EQUATIONS:

To compute the Next State logic (Q3+, Q2+, Q1+, Q0+) and the output control lines for all signals, Karnaugh Maps (K-Maps) were mapped using Sum-of-Products (SOP) criteria. Minimization routines yielded highly optimized Boolean expressions, eliminating redundant literal groupings.

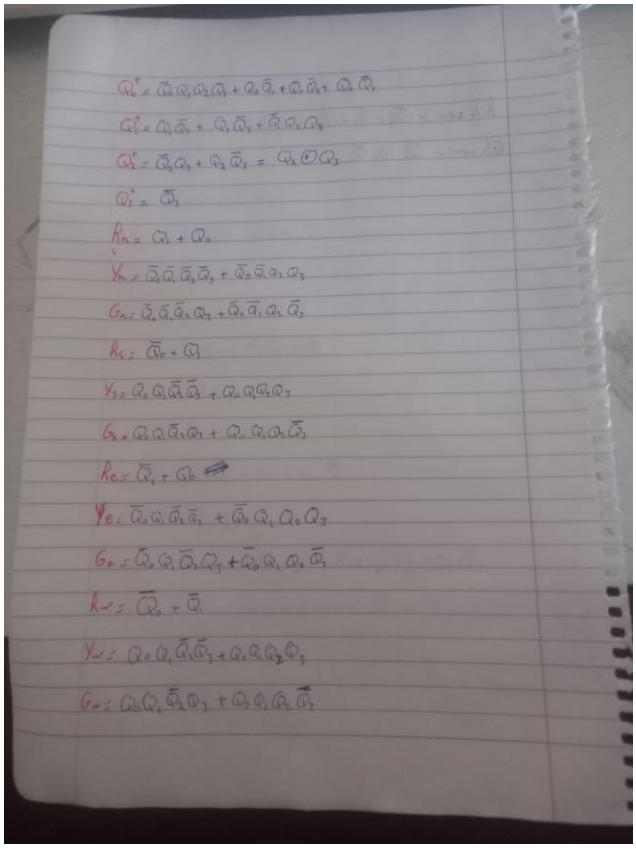


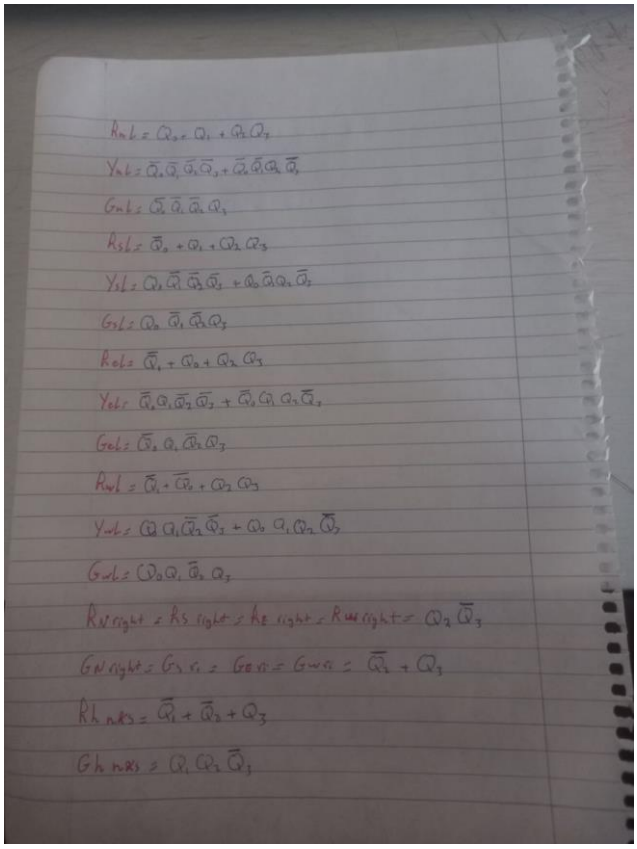


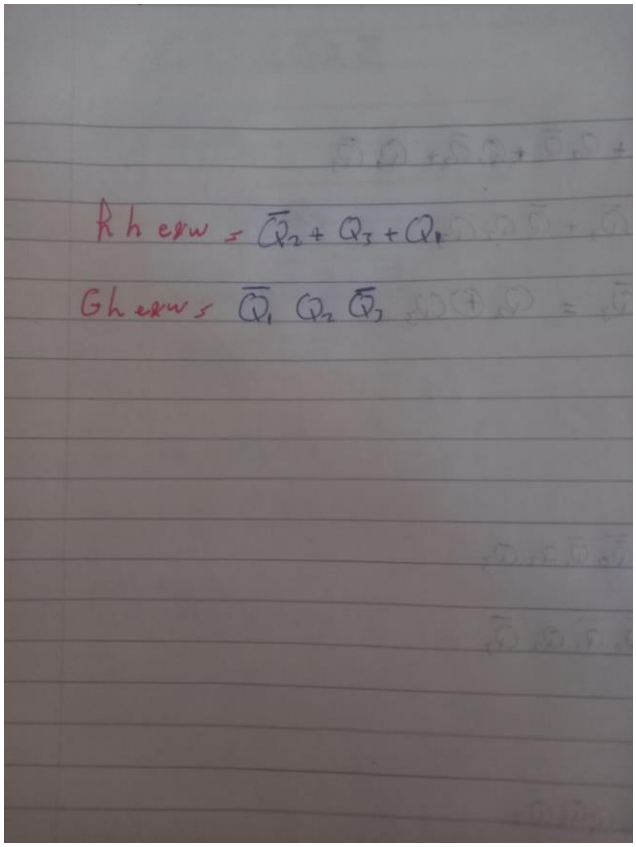




EQUATIONS:







The link of the Logisim File:

https://drive.google.com/file/d/1E1j1dN_IHC9KfkTiTi1BDGf2U_xSQjqK/view?usp=sharing

The Link of the YouTube Video:

https://youtu.be/uEf7u96kMqM

EXCEL Sheet for the truth table:

https://docs.google.com/spreadsheets/d/1j178gHk0-fSWQbBlLY-iEZ6IlC_AmVAo/edit?usp=drive_link&ouid=106478600681951721498&rtpof=true&sd=true

ISLAMIC UNIVERSITY OF GAZA
FACULTY OF ENGINEERING
DEPARTMENT OF ARTificiaL Intelligence ENGINEERING

END OF REPORT

Smart Infrastructure Traffic Control System Project Completed



$$G_{right} = \overline{Q_2} + Q_3$$

Thank you for reviewing.

TECHNICAL ENGINEERING REPORT | PORTFOLIO ASSET

DATE: JUNE 2026